

Mapping a Landslide Event on Puthumala, Kerala, India using SAR Interferometry

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Abstract — Landslides are disturbances in the landscapes causing changes in slope profile in a positive or negative trend induced by natural or anthropogenic activity. Landslides in populated regions could be disastrous and lead to enormous casualties and substantial economic loss. This paper aims to map slow-moving ground displacements with millimetre precision using non-invasive, high resolution, wide area coverage, and cost effective earth observation data. This study uses the phase differences in multi-temporal SAR images with Synthetic Aperture Radar Interferometry (InSAR) for our analysis. Persistent Scatterer Interferometry (PS-InSAR), which utilizes a long time series of radar images was considered to detect the potential coherent measurement points (deformations) in the region. This research work also investigates the efficacy of the StaMPS/MTI toolkit for measuring the surface deformation over the Puthumala region using time series SAR data (29 SLC images) for the period 2019 to 2022.

Keywords — Landslide, SAR Interferometry, PS InSAR, Sentinel data, StaMPS/MTI

I. INTRODUCTION

Landslides are the movement of a mass of rock, debris or earth down a slope. In the past few decades, remote sensing technology has been widely employed in geohazard investigations and assessments using time and weather independent SAR images. The Multi-temporal Interferometric Synthetic Aperture Radar (InSAR) has been proven to be an effective method for monitoring landslides and aims to map slow-moving ground displacements with millimetre precision using non-invasive, high resolution, wide area coverage, and cost effective earth observation data [1]. The first study of InSAR for landslides began in 1999, when Ferretti introduced the permanent scatterer interferometry (PSI) strategy to establish coherence stability to identify landslides over Ancona, Italy [3-4]. In less than two decades, InSAR has become an important information source for landslide deformation studies [5-7] especially for slow to very slow landslides (0–1.6 m/year). Most of these landslides are generally monitored with coherent scatterers that exhibit consistent scattering behavior with time [8]. With the emergence of advanced techniques such as Synthetic Aperture Radar Interferometry (InSAR) and Permanent Scatterer Interferometry (PSI), minor ground deformation with millimetre accuracy is possible using high-resolution microwave data. Persistent Scatterer Interferometry (PS-InSAR), which utilizes a long time series of radar images

was considered to detect the potential coherent measurement points (deformations) in the region [9]. This paper attempts to study the surface deformation that happened due to the landslide disaster accompanied by massive flood at Puthumala, Wayanad, Kerala on 8th August 2019 using SAR Interferometry and PSInSAR techniques.

II. OBJECTIVES

The following are the objectives of our study:

- To generate an interferogram using Sentinel-1 C band Interferometry Wide Swath mode data in VV (Vertical transmit and Vertical receive) polarization.
- To quantify the vertical displacement from the unwrapped interferometric phase information (differential SAR Interferometry.)
- To detect the changes using Normalized Difference Vegetation Index (NDVI), Land use land cover (LULC), and brightness Index.
- To determine land surface movements at a finer scale by using permanent/persistent scatterer interferometric synthetic aperture (PS-InSAR) technique using the StaMPS (Stanford method for persistent scatterers) techniques.

III. STUDY AREA DESCRIPTION

Puthumala is a plantation village in (Muttil) Vythiri taluk of Wayanad district, Kerala, India located in the westerly slope of Western Ghats at an altitude of 1230 m above the mean sea level.

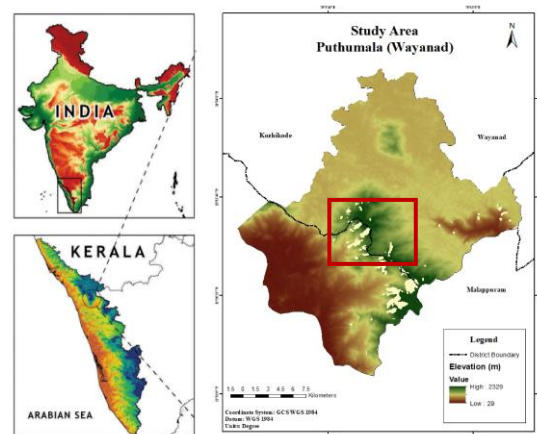


Fig. 1. Study Area Location Map

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The geographic location of the study area ranges from 110 43' to 110 55' N latitudes and 760 05' to 760 20'E longitudes for an area of 69.68 km² (Fig. 1). In general, the elevation is high in the north and northwest and low in the south. Puthumala belongs to the ecologically sensitive area of Western Ghats, and has been categorized under landslide-prone zones of Wayanad. The topography includes attributes ranging from rugged high terrain to flood plains with tropical monsoon climate with 80% rainfall received during southwest monsoon season.

IV. MATERIALS AND METHODS

A. Data Used

Sentinel 1 microwave satellite image and Sentinel-2 multispectral image were obtained from European Radar Observatory for Copernicus (ESA) and preprocessed in Google Earth Engine (GEE) for our analysis.

Table 1. Specifications of Sentinel-1 data

Type	Master	Slave
Date of Acquisition	April 5th 2019	August 22nd 2019
Satellite Type	S1A	S1B
Orbit Direction	Descending	Descending
Polarization	VV+VH	VV+VH
Instrument	SAR-C L1	SAR-C L1
Product Type	SLC	SLC
Sensor Mode	IW	IW
Sub-Swath	IW3	IW3
Orbit file	6-9	6-9
Track	165	165
Cycle Number	553	553

Table 2. Specifications of Sentinel-2 data

Types	Pre landslide	Post landslide
Date of Acquisition	April 27th 2019	September 4th 2019
Data	Multi-spectral	Multi-spectral
Swath Width	290 km	290 km
Band Specifications used for study		
Bands	Central Wavelength (um)	Resolution(m)
Band 2- Blue	0.490	10
Band 3- Green	0.560	10
Band 4- Red	0.665	10
Band 5- Vegetation Red Edge	0.705	20
Band 6- Vegetation Red Edge	0.740	20
Band 7- Vegetation Red Edge	0.783	20
Band 8- NIR	0.824	10
Band 8A- Vegetation Red Edge	0.865	20
Band 10- SWIR- Cirrus	1.375	60
Band 11- SWIR	1.610	20

B. Methodology

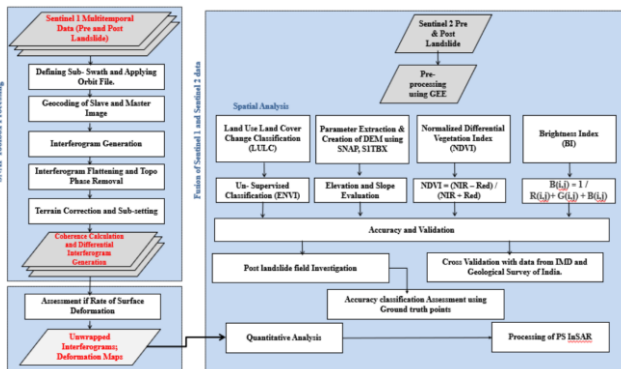


Fig. 2. Schematic representation of adopted methodology of Interferogram Production and Validating Indices

The schematic representation of the methodology used in this study is depicted as Fig. 2. and have three major components: (a) SAR Interferometry processing, (b) Validation using various multispectral indices (NDVI, LULC and Brightness Index), and (c) time series PS-InSAR for ground deformation analysis and validation through field survey on post disaster event.

Interferogram Formation and Coherence Estimation: The complex conjugate of the "slave" and the "master" images (Table 1) are cross-multiplied to create the interferogram, and both images' amplitudes are multiplied while their respective phases are discrete. We used Sentinel-1 Toolbox (S1TBX) which is a collection of processing tools that includes co-registration of images, TOPSAR split, visualization of interferometric phase, deburst and thermal phase removal, multi looking and phase filtering of defined single sub-swath and polarization images in our analysis.

$$\phi = \phi_{DEM} + \phi_{flat} + \phi_{disp} + \phi_{atm} + \phi_{noise} \quad (1)$$

Fusion of Sentinel 1 & Sentinel 2 Images: We have used pre-landslide and post-landslide Sentinel 2 images (Table 2) to fuse with Sentinel-1 images. The process combines different images using specific algorithms to gain better information of the region. Band 2 to 9, 11, 12 from Sentinel-2 and VV, VH bands from Sentinel 1 (IW1) have been layer stacked together using ENVI.

Validation using Multi-spectral Indices: Normalised Difference Vegetation Index (NDVI) characterize the vegetation cover to authenticate the flushed-out area in landslide."(2)"

$$ndvi = \frac{red - NIR}{red + NIR} \quad (2)$$

Unsupervised K-mean classification method was used for LULC classification to discriminate the disaster-affected land use and the changed pattern. Brightness Index (BR) was calculated using the visible bands of the multispectral image since it is influenced by the soil moisture, presence of salt and organic matters and density of vegetation cover. In this analysis, the brightness index would be helpful in identifying the exposure of barren soil due to the loss of vegetation."(3)"

$$BR_{(i,j)} = \frac{1}{R_{(i,j)} + G_{(i,j)} + B_{(i,j)}} \quad (3)$$

Stanford Method for Persistent Scatterer Interferometry (StaMPS). The StaMPS approach [9] was developed to detect a group of phase stable pixels called Persistent Scatter (PS) by using the spatial correlation of the phase of the 39 pixels in a sequence of interferograms. An initial filter is applied based on amplitude dispersion, which is kept high to identify a larger number of phase stable pixels. The phase stability of the selected pixels is evaluated, and the pixels that are eventually selected as PS are used to estimate the deformation; the procedure is detailed in following sections. Instead of employing an assumed model, the approach characterize the temporal model of deformation, eliminating the necessity for an a priori temporal model.[10]

V. RESULTS AND DISCUSSION

A. Interferometry to identify the Surface Deformation

In the study of Puthumala Landslide, for SAR interferometry out of the two images, the master image used was acquired on 5th April 2019 and the slave image used was acquired on 22th August 2019. Interferometry have been applied to the spitted tiles; thus, surface deformation has been detected (Fig. 3) and indicated with different colours. Fig 3(a) shows a blue colour with a negative subsidence value. The slope of the region has been pointed towards the west. Fig. 3(b) is the region which carries the debris of the slide as indicated in the colour red and green with a positive value. In the case of Fig 3(c), it has been recorded as -1.493 in the majority of the parts, where the accumulation of the flowed debris has concentrated in surrounding regions. Now, moving on to the landslide-affected areas of Puthumala, Fig 3(d) gives a closer detailing of the minute but very drastic changes of value within a small portion of land. Gradation has happened very rapidly with -2.342 (slide) and suddenly to 1.957 (accumulation).

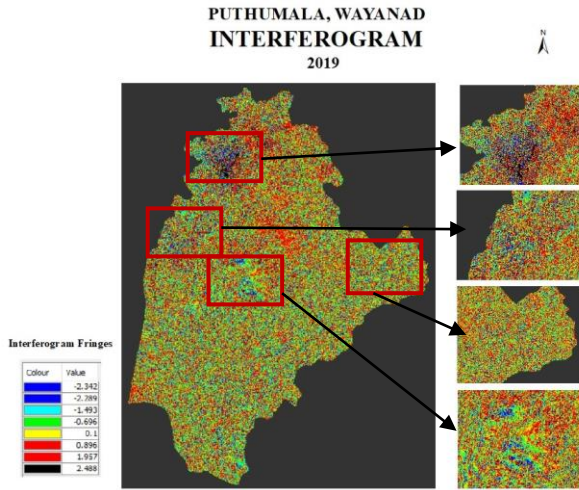


Fig. 3. The map of phase shifts, or interferogram, is depicted with a repeating colour scale that shows relative displacement between the first and the second acquisition. The direction of displacement- subsidence or uplift- is indicated by sequence of the colour progression of the fringe(s) toward the centre of the deforming feature

B. Validation using NDVI

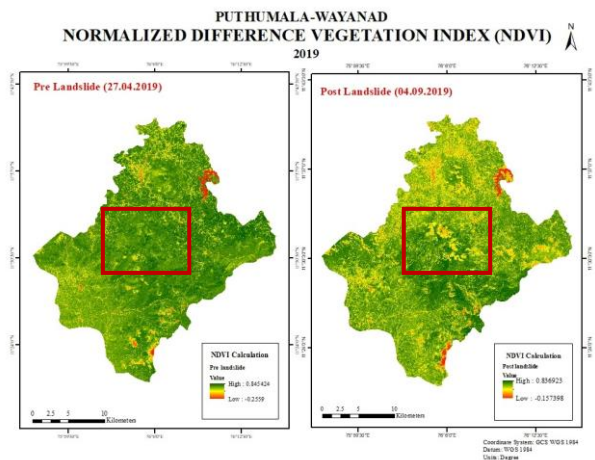


Fig. 4. Normalized Differential Vegetation Index for Puthumala region before and after landslide

The change in vegetative cover has been computed and analyzed using NDVI using the fused image of Sentinel 1

(Microwave data) and Sentinel 2 (Multispectral data). The pre- and post-landslide NDVI change is presented in Fig. 4. In the pre-landslide section the region was covered by thick vegetation, which can be quantified between 0.667 to 0.845 as shown in the variation index of NDVI calculation and the same area post landslide has a value of 0.374 as per the index.

C. LULC Change Detection

The LULC cover change map for the landslide event is given as Fig. 5. The change detection map depicts a clear wash-off of the vegetation/ forest area (indicates the plantation in this region) and turns it into barren land and depressions.

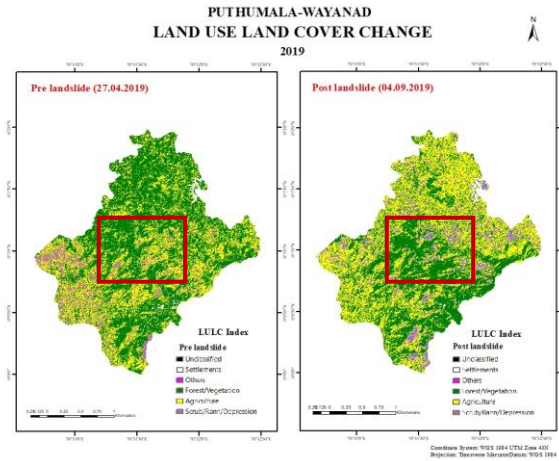


Fig. 5. LULC change detection map for pre- and post-landslide in Puthumala, Kerala

D. Change in Brightness Index

The change in the brightness index between the pre-and post-landslide event was calculated and given as Fig. 6. High brightness index refers to the exposure to the barren land, and the low brightness index shows thick vegetation cover.

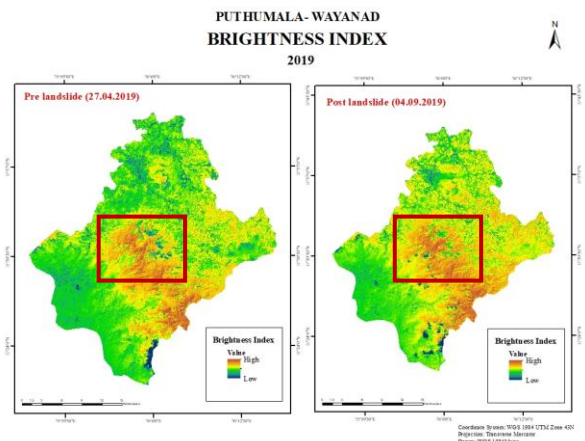


Fig. 6. Change in brightness index for Puthumala region before and after landslide

E. Persistent Scatterer (PS) InSAR analysis for Deformation Measurements

Using 29 C-Band SAR SLC images, 28 interferograms were formed. Using the Amplitude Dispersion (D_A) and coherence (γ) value of 0.4 and 0.005 respectively, 1222 PS pixels were finally detected. The PS displacement plot shown in Fig. 8 shows a times series (TS) surface displacement in the range of -4.7 to 4.6 mm per year, in which positive sign and the negative sign shows

displacement, with warm and cold colours showing displacement, towards and away from the satellite LOS respectively. The X and Y axes shows longitude and latitude respectively [11]. The movement with negative sign indicates the surface affected by subsidence or similar activities. The displacement plot shows time series deformation with respect to the first differential interferogram (Master Image). PS density in few areas is slightly low because of the presence of a majority of pixels having distributed scattering properties. 29 images (from 2019 to April 2022) including the master tile was employed to calculate the accurate and up-to-date velocity of surface movement in mm/year using StaMPS technique. Suddenly after the landslide of Puthumala on 8th August 2019, the displacement rate is shown higher (with red patches). As time moves, the displacement rate has been lowered. The same higher deformation trend can be seen in the yearly monsoon season, where landslides usually continue in unstable terrain. The areas which appear in moderate colour (light green) specify the regions where very little deformation occurred. The results were then verified using post-disaster field visits` conducted in February 2020 (Fig. 9)

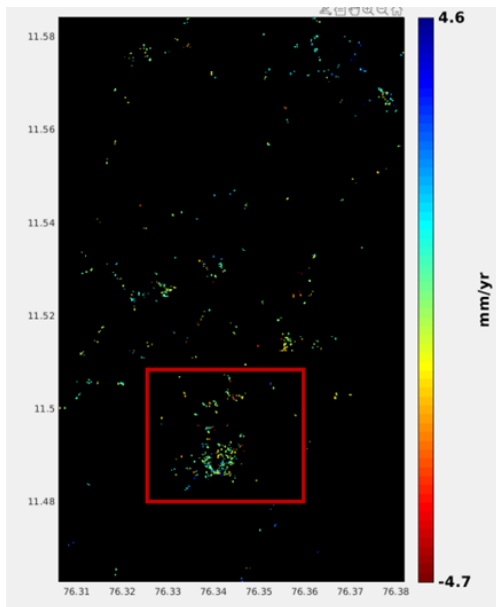


Fig. 8.Velocity of Ground Movement in the study region (in mm/yr)



Fig. 9.Field Survey conducted on Feb 2020

VI. CONCLUSION

The SAR interferometry approach is proven to be useful in the preliminary detection and displacement quantification of landslides. Further, the SAR data allows us for a synoptic analysis of landslides without the constraints of time or

climate. The most straightforward approach for creating a differential interferogram is the 2-pass method we employed in our study. The results of the StaMPS/MTI approach proved to be an efficient method to determine the surface deformation at millimetre precision. The method we used has proved to be effective in understanding the terrain movement, identifying potential landslide risk zone, and consequent prediction of future landslide events of landslide.

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